

TRACK IT

1. Project Description

● **The E-Learning Platform** is an innovative, scalable, and secure digital ecosystem designed to revolutionize online education by delivering personalized learning experiences, advanced AI-driven tools, and seamless integration with career development opportunities. The platform empowers learners, instructors, and institutions to achieve their educational and professional goals through a unified, user-centric approach.

● **Learners** interact with adaptive AI tutors that analyze their performance, preferences, and career aspirations to generate tailored learning paths. The system tracks progress in real time, provides interactive quizzes with AI-powered grading, and offers hands-on coding labs and virtual projects to reinforce skill mastery. Gamification elements like badges, leaderboards, and streak tracking foster engagement and accountability.

● **Instructors** utilize advanced course creation tools, including AI-assisted content generation (e.g., auto-generated quizzes from lecture materials) and multi-format delivery (videos, SCORM packages, live webinars). Version control allows updates without disrupting enrolled students, while DRM and watermarking protect intellectual property. Instructors can monetize content through subscriptions, one-time purchases, or employer-sponsored courses, with automated financial reporting and payout management.

● **Enterprise-Grade Security** ensures compliance with global standards (GDPR, CCPA) through multi-factor authentication (MFA), OAuth logins, and role-based access control. Suspicious activity detection, session expiration policies, and blockchain-verified certifications safeguard user data and content integrity.

● **Career Integration** bridges learning with job opportunities through AI-driven resume builders, skill gap analysis, and employer partnerships. The platform matches learners with internships and roles based on completed courses, while companies can sponsor courses aligned with in-demand skills.

● **A premium subscription**, accessible via integrated payment gateways (PayPal, Stripe), unlocks advanced features like live mentorship sessions, proctored exams, and priority access to hackathons. Institutions and employers receive dedicated dashboards to track learner progress, compliance, and ROI.

● **Financial Transparency:** The platform deducts a **15% service fee** from all transactions (course purchases, subscriptions). Instructors and mentors receive automated payouts for their earnings, with real-time tracking of revenue and deductions. Users view itemized billing for subscriptions, certifications, and mentorship sessions, ensuring clarity in all transactions.

2. Team Members

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3. Initial Customer Requirements

1. Secure user authentication with MFA and OAuth.
2. AI-driven course recommendations and adaptive learning paths.
3. Integration with PayPal/Stripe for payments and subscriptions.

4. Real-time progress tracking and blockchain-based certifications.
5. DRM-protected content and plagiarism detection.

Costumer Requirements

1. User Authentication & Security

- The system shall allow users to register and log in using email, password, and social media accounts.
- The system shall support **Multi-Factor Authentication (MFA)** and OAuth-based login.
- The system shall manage user sessions securely, with automatic **session expiration policies** based on user roles.
- The system shall enforce **data privacy regulations (GDPR, CCPA, etc.)** and allow users to export or delete their data.
- The system shall implement **suspicious activity detection & session revocation** for security.

2. User Roles & Permissions

- The system shall implement **Role-Based Access Control (RBAC)** with custom permissions for students, instructors, and administrators.
- The system shall allow users to create and manage personal **profiles, discussion posts, and activity feeds**.
- The system shall maintain **audit logs & security tracking** for user actions and data access.

3. Payments & Monetization

- The system shall integrate with multiple **payment gateways (PayPal, Stripe, etc.)** for seamless transactions.
- The system shall support **subscription-based plans, one-time purchases, and tiered access**.
- The system shall enable administrators to create and manage **discounts, promo codes, and referral rewards**.

- The system shall generate **financial reports & manage payouts** for instructors.
- The system shall support **in-app purchases & wallet-based transactions**.

4. Course Management & Delivery

- The system shall allow instructors to **create, edit, and organize courses** with structured modules.
- The system shall track student progress in real-time and provide **adaptive difficulty settings** for assessments.
- The system shall support **AI-assisted course creation** (e.g., auto-generating quizzes from content).
- The system shall provide **multi-format content delivery** (videos, PDFs, SCORM packages, interactive media).
- The system shall issue **certifications** upon course completion, with public verification links.
- The system shall support **course versioning**, allowing instructors to update content without affecting enrolled students.

5. Learning Experience & Interaction

- The system shall support **self-paced and instructor-led learning** with scheduled assignments.
- The system shall provide **interactive quizzes, assessments, and AI-powered grading** for subjective responses.
- The system shall offer **hands-on coding exercises, virtual labs, and project-based learning**.
- The system shall include **social learning features**, such as discussion forums and student-to-student messaging.

6. AI-Powered Learning & Personalization

- The system shall implement **adaptive learning paths** based on student performance and preferences.
- The system shall provide **AI-based course recommendations** tailored to career goals and interests.
- The system shall conduct **skill gap analysis & suggest learning paths** for users.
- The system shall use **predictive analytics to prevent dropouts & improve retention**.
- The system shall support **AI tutors & chatbot assistants** for 24/7 guidance.

7. Gamification & Engagement

- The system shall include **points, badges, and leaderboards** to incentivize learning.
- The system shall issue **achievements, digital certificates, and level-based progressions**.
- The system shall introduce **seasonal challenges, streak tracking, and limited-time events**.
- The system shall enable **study groups, collaborative projects, and peer mentoring**.

8. Live Sessions & Webinars

- The system shall support **real-time webinars & live lectures** with integrated video conferencing tools.
- The system shall provide **interactive Q&A, polls, and chat moderation** during live sessions.
- The system shall allow students to access **recorded sessions & playback with chapter markers**.
- The system shall enable **AI-powered live transcriptions & translations** for accessibility.

9. Hands-On Projects & Assessments

- The system shall include **real-world case studies & capstone projects** in courses.
- The system shall support **project submissions, peer reviews, and instructor feedback**.
- The system shall allow students to **integrate their coursework into a digital portfolio**.
- The system shall implement **proctored exams, plagiarism detection, and AI-assisted grading**.

10. Course Discovery & Recommendations

- The system shall provide **trending and personalized course recommendations** based on user behavior.
- The system shall offer **structured learning paths & skill progression roadmaps**.
- The system shall include **advanced search & filtering with natural language processing (NLP)**.

11. Career Support & Job Matching

- The system shall provide a **Resume & CV Builder** with AI-powered formatting suggestions.
- The system shall implement **AI-based job & internship matching** based on student skills & completed courses.
- The system shall allow **employers to create job listings & search for talent**.
- The system shall offer **company-sponsored courses linked to job opportunities**.

12. Mentorship & Industry Networking

- The system shall support **AI-based mentor-mentee matching** based on career goals.
- The system shall enable **one-on-one & group mentorship** through video calls.
- The system shall integrate **networking & industry events**, including career fairs.

13. Hackathons & Competitions

- The system shall host **coding challenges, hackathons, and leaderboards**.
- The system shall support both **solo & team participation**.
- The system shall provide **rewards, job opportunities, & employer-sponsored challenges**.

14. Roadmap & Skill Progression

- The system shall offer **customizable learning roadmaps** with career-specific tracks.
- The system shall track **prerequisites & suggest next steps** based on user progress.
- The system shall provide **visual progress tracking & milestone achievements**.

15. Admin Dashboard & Platform Management

- The system shall provide an **admin dashboard** for managing users, courses, & **financials**.
- The system shall include **system health monitoring & real-time analytics**.
- The system shall maintain **audit logs, compliance tracking, & user activity reports**.

16. Mobile Learning & Offline Access

- The system shall support **cross-platform learning** (iOS, Android, Web, PWA).
- The system shall allow **offline course downloads** with progress syncing.
- The system shall enable **push notifications** for reminders & study alerts.

17. LMS & Third-Party Integrations

- The system shall provide **APIs** for integration with external LMS platforms.
- The system shall support **Single Sign-On (SSO)** & OAuth authentication.
- The system shall enable **webhooks** & third-party app integrations (Slack, Zapier, Trello, LinkedIn, etc.).

18. Content Security & Intellectual Property Protection

- The system shall implement **Digital Rights Management (DRM)** & watermarking.
- The system shall support **content integrity verification** & access restrictions.
- The system shall prevent **unauthorized sharing** & screen recording for premium content.

19. AI & Automation Features

- The system shall provide **smart notifications** & automated study reminders.
- The system shall support **automated feedback** on assignments.
- The system shall include **AI-powered resume optimization** & job application assistance.

20. Certification & Accreditation System

- The system shall support **stackable credentials** & micro-degrees.
- The system shall implement **blockchain-based certificates** for verification.
- The system shall ensure **employer recognition** & job integration.

Subsystem Table

Subsystem Name	Subsystem Function	Subsystem Interface

User Authentication & Security	• Function: Ensures secure user authentication and session management. • Features: User registration, login, multi-factor authentication (MFA), OAuth integration, secure session handling, and data encryption.	<pre>public interface AuthenticationService { boolean registerUser(String email, String username, String password); boolean loginUser(String emailOrUsername, String password); boolean resetPassword(String email); boolean updatePassword(String userId, String oldPassword, String newPassword); boolean enableMFA(String userId); boolean verifyMFA(String userId, String code); void logoutUser(String userId); }</pre>
User Roles & Permissions	• Function: Manages user roles, access control, and security policies. • Features: Role-based access control (RBAC), user permissions management, audit logs, and activity tracking.	<pre>public interface RoleManagement { boolean assignRole(String userId, String role); List<String> getUserPermissions(String userId); boolean checkPermission(String userId, String action); List<String> getAuditLogs(); }</pre>
Payments & Monetization	• Function: Handles payments, subscriptions, and revenue optimization. • Features: Payment gateway integration, subscription management, discount handling, financial reporting, and fraud detection.	<pre>public interface PaymentService { boolean processPayment(String userId, double amount, String paymentMethod); boolean initiateSubscription(String userId, String planId); boolean cancelSubscription(String userId); boolean isSubscriptionActive(String userId); List<String> getPaymentHistory(String userId); }</pre>

Course Management & Delivery	Function: Facilitates course creation, organization, and content delivery. Features: Course structuring, multimedia content support, version control, content updates, and certification issuance.	<pre>public interface CourseService { boolean createCourse(String instructorId, String title, String description); boolean updateCourseContent(String courseId, String moduleId, String content); List<String> getCourseModules(String courseId); boolean publishCourse(String courseId); }</pre>
Learning Experience & Interaction	Function: Enhances user engagement with interactive learning tools. Features: Self-paced learning, instructor-led courses, quizzes, assessments, and gamification elements.	<pre>public interface LearningExperience { boolean enrollInCourse(String userId, String courseId); String trackProgress(String userId, String courseId); boolean completeLesson(String userId, String lessonId); }</pre>
AI-Powered Learning & Personalization	Function: Adapts learning content dynamically based on user performance. Features: AI-driven course recommendations, personalized learning paths, skill gap analysis, and predictive analytics.	<pre>public interface AIRecommendationService { List<String> getRecommendedCourses(String userId); String analyzeSkillGap(String userId); boolean adjustLearningPath(String userId); }</pre>
Gamification & Engagement	Function: Increases motivation and retention using game mechanics. Features: Points, badges, leaderboards, achievement tracking, and social learning groups.	<pre>public interface GamificationService { boolean awardBadge(String userId, String badgeId); int getUserPoints(String userId); List<String> getLeaderboard(); }</pre>
Live Sessions & Webinars	Function: Enables real-time learning through live instructor sessions. Features: Live streaming, interactive Q&A, recorded sessions, student participation tracking, and poll integration.	<pre>public interface LiveSessionService { boolean scheduleLiveClass(String instructorId, String courseId, String dateTime); boolean recordSession(String sessionId); String getRecordedSession(String sessionId); }</pre>

Hands-On Projects & Assessments	• Function: Provides experiential learning through projects and assessments. • Features: Real-world case studies, project submission, instructor reviews, AI-based assessments, and plagiarism detection.	<pre>public interface AssessmentService { boolean submitProject(String userId, String projectId, String submissionUrl); String getInstructorFeedback(String projectId); boolean verifyPlagiarism(String userId, String submissionId); }</pre>
Course Discovery & Recommendations	• Function: Helps users find relevant courses based on interests and learning trends. • Features: Personalized recommendations, trending courses, roadmap generation, and adaptive search filters.	<pre>public interface CourseDiscoveryService { List<String> getTrendingCourses(); List<String> getPersonalizedRecommendations(String userId); List<String> searchCourses(String keyword, String category, String difficulty); }</pre>
Certification & Accreditation System	• Function: Issues and verifies certifications to validate learning achievements. • Features: Stackable credentials, blockchain verification, employer recognition, and global accreditation.	<pre>public interface CertificationService { boolean generateCertificate(String userId, String courseId); boolean verifyCertificate(String certificateId); boolean shareCertificate(String userId, String courseId, String platform); }</pre>
Career Support & Job Matching	• Function: Bridges the gap between education and employment opportunities. • Features: Resume builder, job and internship matching, employer connections, and AI-driven job recommendations.	<pre>public interface CareerSupportService { String generateResume(String userId, String templateId); List<String> getJobRecommendations(String userId); boolean trackJobApplication(String userId, String jobId); }</pre>

Mentorship & Industry Networking	• Function: Connects learners with industry experts and mentors. • Features: AI-based mentor matching, one-on-one mentorship sessions, networking events, and career growth tracking.	<pre>public interface MentorshipService { String matchMentorMentee(String menteeId); boolean scheduleMentorshipSession(String mentorId, String menteeId, String dateTime); boolean joinNetworkingEvent(String userId, String eventId); List<String> getAvailableMentors(String industry); }</pre>
Hackathons & Competitions	• Function: Encourages skills development through competitive coding challenges. • Features: Hackathons, leaderboards, coding challenges, industry partnerships, and prize-based competitions.	<pre>public interface HackathonService { boolean createHackathon(String eventName, String problemStatement); boolean registerForHackathon(String userId, String hackathonId); Map<String, Integer> getLeaderboard(String hackathonId); boolean awardPrize(String userId, String prizeId); }</pre>
Roadmap & Skill Progression	• Function: Guides users through structured learning paths. • Features: Custom learning roadmaps, prerequisite tracking, skill assessments, and milestone tracking.	<pre>public interface RoadmapService { String createLearningPath(String userId, String careerGoal); boolean addCourseToPath(String pathId, String courseId); boolean trackProgressInRoadmap(String userId, String pathId); }</pre>
Admin Dashboard & Platform Management	• Function: Provides administrators with complete control over platform operations. • Features: User management, system monitoring, analytics dashboard, compliance tracking, and content moderation.	<pre>public interface AdminService { boolean manageUserAccount(String adminId, String userId, String action); boolean updateSystemSettings(String settingName, String value); List<String> getSystemLogs(); }</pre>

Mobile Learning & Offline Access	• Function: Enables seamless learning across devices with offline support. • Features: Cross-platform support (iOS/Android), offline content downloads, progress syncing, and push notifications.	<pre>public interface MobileLearningService { boolean enableMobileAccess(String userId); boolean downloadCourseContent(String userId, String courseId); boolean syncOfflineProgress(String userId); }</pre>
LMS & Third-Party Integrations	• Function: Ensures compatibility with external learning systems and APIs. • Features: External LMS integration, API access, Single Sign-On (SSO), and webhook event-based updates.	<pre>public interface LMSIntegrationService { boolean connectToExternalLMS(String lmsPlatform); boolean syncUserData(String userId, String courseId); boolean enableSSO(String provider); }</pre>
Content Security & Intellectual Property Protection	• Function: Safeguards intellectual property and prevents unauthorized content usage. • Features: Digital rights management (DRM), watermarking, role-based content restrictions, and security monitoring.	<pre>public interface ContentSecurityService { boolean applyDRM(String contentId); boolean addWatermark(String contentId, String userId); boolean restrictContentAccess(String userId, String role); }</pre>
AI & Automation Features	• Function: Enhances efficiency using AI-powered automation and analytics. • Features: Smart notifications, automated assignment grading, resume optimization, and engagement tracking.	<pre>public interface AIService { boolean sendSmartReminder(String userId, String reminderType); String generateAIFeedback(String userId, String assignmentId); List<String> getUserEngagementInsights(String userId); }</pre>

Traceability Matrix

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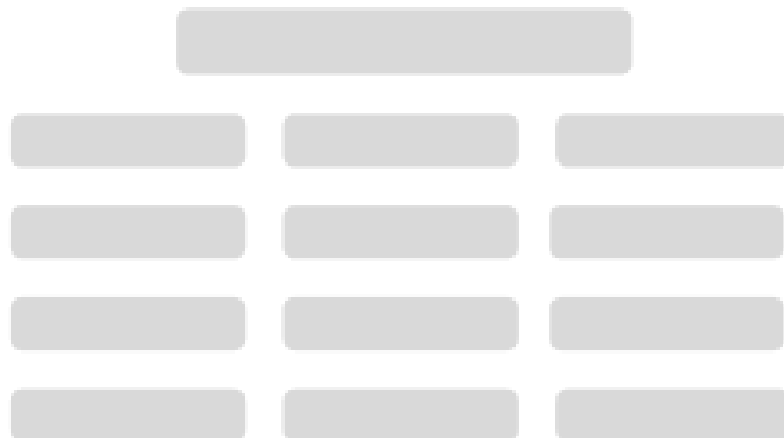
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TrackIT Road Map

TrackIT is a Community Effort To Create Roadmaps, Guides And Other Educational Content To Help Guide Developers In Picking Up A Path And Guide Their Learning.

Design and Development



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Track 1

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03
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Building a Professional Website

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02
User Research and Wireframing

03
Prototyping and Usability Testing

04
Design Fundamentals

05
Building a Professional Website

Mobile App Development

Learn the fundamentals of mobile app development, including user research, wireframing, and prototyping. This course is designed to help you learn the skills you need to create user-friendly mobile apps.

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Introduction to Mobile App Development

02
User Research and Wireframing

03
Prototyping and Usability Testing

04
Design Fundamentals

05
Building a Professional Website

Graphic Design for Beginners

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02
Color Theory and Typography

03
Layout Design and Responsive Design

04
Design Fundamentals

05
Building a Professional Website

Front-End Web Development

Learn the fundamentals of front-end web development, including HTML, CSS, and JavaScript. This course is designed to help you learn the skills you need to create professional-looking websites.

View Course



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01
Introduction to Front-End Web Development

02
HTML and CSS Fundamentals

03
JavaScript Fundamentals

04
Building a Professional Website

05
Building a Professional Website

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